

Challenges (1): study design

Study designs may vary by stage of outbreak/public health need

Table 2: Viral kinetics surveillance studies and priorities during an outbreak			
Stage of outbreak	Early: first few hundred cases; incidence low but increasing.	Middle: sustained transmission with high incidence; possibly covering multiple epidemic waves.	Late: non-emergency phase; majority of the population has likely been exposed to the pathogen and/or vaccinated; low incidence.
Research priorities	Key epidemiological parameters such as mean and variance of infection duration; effect of demographic covariates such as age; generate testable hypotheses for infectiousness, immunity and treatment.	Viral kinetics estimates for different subpopulations; impact of treatment, symptom severity, immunity, and strains/variants on viral kinetics parameters; more challenging biological questions such as identifying correlates of infectiousness or explaining unusual kinetics such as rebounds; embedding in epidemiological models to simulate interventions.	Kinetics of reinfection; effect of new strains/variants and infection history on typical viral trajectories; stability of kinetics trajectories as pathogen becomes endemic (e.g., what is the typical clearance rate in a highly immune population?).
Study design	Adaptive study designs to maximize recruitment and testing as more is learned about the pathogen; likely recruiting unrepresentative patients following symptom onset, hospitalization or through contact tracing; likely frequent testing until optimal sample sizes can be determined.	Household or exposure-based studies aiming to recruit participants as early in infection as possible; representative study population including risk groups and children if ethical; minimal longitudinal testing to power viral kinetics models; initiate sub-studies to address key unknowns identified during the early phase.	Low prevalence means that prospective study designs will likely be inefficient; unlikely to justify intensive case finding; return to testing hospitalised patients, outpatients or opportunistic recruitment.
Modelling needs	Simple statistical models to share key parameters quickly and in an accessible way; prototype mechanistic models to highlight key unknowns and guide future studies e.g., therapeutic target product profile.	Sophisticated statistical models to continue reporting key parameters; hierarchical models to quantify impact of key covariates such as age on viral kinetics parameters; stable, minimal mechanistic models to address key biological questions such as estimating within-host R_0 , the role of different immune compartments, mechanism behind atypical trajectories.	Mechanistic models are likely to be more useful than simple phenomenological models, as there will be fewer detected infections (smaller study population) but improved technology and understanding to measure different compartments per individual (better sample size per individual).
Reporting	Pre-prints; press releases; data sharing agreements.	Regular (weekly) and consistent reporting, similar to epidemiological surveillance reports; pre-prints; posting data and analyses to dashboards or public repositories.	Publications; infrequent epidemiological surveillance reports; public data repositories.

Challenges (2): data storage and access

Creating a global viral kinetics library

A library of standardized viral kinetics data with basic model fits/parameter estimates

Complementary to efforts like:

- WHO Global Repository of Epidemic Parameters
- Global Vaccine Library
- NextStrain

Benefits:

- Facilitates comparisons between study designs, pathogens, time frames, demographic groups...
- Enables quick dissemination of data
- Enables closer collaboration across the data collection/modeling/policy design spectrum

Challenges:

- Balancing data comparability with completeness
- Ensuring data quality
- Integrating robust + flexible modeling tools
- Ensuring appropriate credit (see next slide!)

The screenshot shows the 'Filters' section of the Viral Kinetics Explorer. It includes an 'Age Range' slider set from 0 to 100, a 'Study ID' field with 'ke2022' and 'kissler2023' selected, and a 'Sample Type' field with 'nasal', 'saliva', and 'nasal_orophary...' selected.

Viral Kinetics Explorer

Upload your viral kinetics CSV (or use default below)

Drag and drop file here
Limit 200MB per file • CSV

Browse files

Using default viral kinetics dataset.

Filtered Data Preview

	StudyID	PersonID	InfectionID	SampleID	TimeDays	Symptoms1	Symptoms2	Symptoms3	Symptoms4	Comorbidity1	Comorbidity2	Comorbidity3	Comorbidity4	T
2	ke2022	443108	None	None	-2	None	None	None	None	None	None	None	None	
3	ke2022	443108	None	None	-1	None	None	None	None	None	None	None	None	
4	ke2022	443108	None	None	0	None	None	None	None	None	None	None	None	
5	ke2022	443108	None	None	1	None	None	None	None	None	None	None	None	
6	ke2022	443108	None	None	2	None	None	None	None	None	None	None	None	
7	ke2022	443108	None	None	3	None	None	None	None	None	None	None	None	
8	ke2022	443108	None	None	4	None	None	None	None	None	None	None	None	
9	ke2022	443108	None	None	5	None	None	None	None	None	None	None	None	
10	ke2022	443108	None	None	6	None	None	None	None	None	None	None	None	
11	ke2022	443108	None	None	7	None	None	None	None	None	None	None	None	

